

2.3.4

The cyclical nature of trigonometric functions lead to a 2d representation of a 3d cylinder. When looking at the edges of the eye-like shape, the corners represent the same mathematical value. In the trigonometric circle, $\pi/2$ and $-\pi/2$ are the same value. In the linear representation of the axis, they are separate whilst in reality the axis just curves around.

2.5

The chaotic nature of the Lorenz system can be seen through the shape of the function of the solution. The deterministic chaos defined by his theory shows that with a small change in the initial conditions there is a large effect on the Lorenz attractor. It is considered deterministic as the outcome can be predicted if the parameters are precise enough, but because of the nature of chaos, even a small variation in them will change the outcome leading to large inaccuracies in predictions in the long-term.

2.6.1 and 2.6.2

We believe that we can approximate the area of the phase space function with the integration of the two conjoint hyperboles. In other words, the hyperbolic ellipses shape can simply be integrated using the formula: $\int_a^b f(x) - g(x) dx$ where $f(x)$ is larger than $g(x)$ and both are hyperboles. Because of Liouville's theorem, the approximation should approximate all the areas that the function can take.

Because of the nature of Euler's method, it is but an approximation the function will not form a closed shape which might lead to major inaccuracies.